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Nitte (Deemed to be University)

(Established under Sec (3) of UGC ACT,1956)

First Semester MCA (CBCS) Degree Examinations, Dec. 2025 - Jan. 2026

25MCA103 - DATA COMMUNICATION AND NETWORKS

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Describe the structure and functioning of the Internet Architecture.	08	L2	1	1
b) Check whether there is any error during transmission using polynomial method for the message X^6+X^2+1 and pattern X^3+X^2+1 .	12	L3	1	2
2. a) Explain with Suitable example to show different type of errors in networks.	10	L3	1	1
b) Explain the working of TCP/IP Architecture with neat diagram.	10	L2	1	1

Unit – II

3. a) Explain the working of DNS in Internet Directory Services. Discuss its database structure and operations.	10	L2	2	1
b) In a 5-frame transmission, the first two frames are successfully received, but the 3rd frame is lost, while the 4th and 5th frames reach the receiver. Explain how this situation is handled using Go-Back-N ARQ and Selective Reject ARQ protocols. Also, consider a case where one of the acknowledgments (ACK) is lost. Illustrate your answer with the help of a suitable diagram.	10	L3	2	2
4. a) Explain Sliding window flow control.	10	L2	2	1
b) Explain the working of the Simple Mail Transfer Protocol with the help of a suitable example of email transfer.	10	L2	2	1

Unit – III

5. a) Explain how the Transport Layer performs multiplexing at the source and demultiplexing at the destination. Illustrate your answer with a suitable example.	08	L2	3	1
b) Compare the TCP Segment with the SCTP Packet. Explain their frame formats with neat diagrams.	12	L3	3	2
6. a) Explain the TCP connection establishment and termination process using the Finite State Machine (FSM) model.	14	L2	3	1
b) Explain the concept of error control in transport layer.	06	L2	3	1

Unit – IV

7. a) Explain the implementation of connectionless service in the Network Layer with diagrammatic approach.	10	L2	4	1
b) How does hierarchical routing enhance network scalability and efficiency? Explain with example.	10	L2	4	1
8. a) What is the concept of store and forward packet switching, and how does it work in data communication networks?	06	L1	4	1
b) Compare virtual circuit with datagram subnet.	10	L3	4	2
c) Write a note on Optimality principle.	04	L2	4	1

Unit – V

9.	a)	Define congestion. Explain its effects.	08	L1	5	1
	b)	Explain Datagram Congestion Control Protocol (DCCP) and compare it with TCP.	12	L2	5	1
10.	a)	Explain various congestion control techniques.	12	L2	5	1
	b)	Discuss different traffic management policies used in networks.	08	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
